RĐ:Đậu Thế Phiệt	Ngày:	PD:Nguyễn Tiến Dũng	Ngày ....
Ký tên		Ký tên	

Đại học Bách khoa-ĐHQG TPHCM Khoa Khoa học Ứng dụng	**THI GIỮA KỲ**	**Kỳ/năm học**	**II**	**2023-2024**
		Ngày thi	**04/08/2024**	
	Môn học	**Môn ĐẠI SỐ TUYẾN TINH**		
	Mã môn học	**MT1007**		
	Thời gian	50 phút	Mã đề	1101

Notes: - Đề thi trắc nghiệm gồm có 40 câu/4 trang.
- *Sinh viên không được dùng tài liệu. Nộp lại đề thi và giấy nháp cho giám thị.*
-*Mỗi câu trắc nghiệm sai: -1/5 số điểm của câu đó. Nếu không khoanh thì không trừ điểm.*

EXAM

(**Question 1 to question 3**)

Let $A = \begin{bmatrix} 2 & 1 & -2 & 1 \\ 3 & 2 & 1 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & m \\ 2 & -2 \end{bmatrix}$, $m \in \mathbb{R}$, be two matrices.

Question 1 (L.O.1, L.O.2). Which of the following statements is CORRECT?

A. $BA = \begin{bmatrix} 3m+2 & 2m+1 & m-2 & 1-2m \\ -2 & -2 & -6 & 6 \end{bmatrix}$. **B.** $A + B = \begin{bmatrix} 3 & m+1 \\ 5 & 0 \end{bmatrix}$.

C. None of the others. **D.** BA does not exist. **E.** $AB = \begin{bmatrix} 8 & 2m-6 \\ 5 & m-4 \\ 0 & -2m-2 \\ -3 & m+4 \end{bmatrix}$.

Question 2 (L.O.1, L.O.2). Let $f(x) = 3x^2 - 2x - 4$ be a polynomial. Find $f(B)$.

A. $f(B) = \begin{bmatrix} 4m-5 & -5m-4 \\ -14 & 4m+10 \end{bmatrix}$.

B. $f(B) = \begin{bmatrix} 4m-5 & -2m-6 \\ -3m-4 & 4m+10 \end{bmatrix}$.

C. $f(B) = \begin{bmatrix} 4m-5 & -2m-10 \\ -3m-8 & 4m+10 \end{bmatrix}$.

D. None of the others.

E. $f(B) = \begin{bmatrix} 6m-3 & -5m \\ -10 & 6m+12 \end{bmatrix}$.

Question 3 (L.O.1, L.O.2). Given that $m = 2$. Find a matrix X such that: $2BX = 3A + X$.

A. $X = \begin{bmatrix} \frac{64}{21} & \frac{34}{21} & -\frac{16}{7} & \frac{6}{7} \\ \frac{68}{21} & \frac{44}{21} & \frac{4}{7} & -\frac{12}{7} \end{bmatrix}$. **B.** $X = \begin{bmatrix} \frac{22}{7} & \frac{13}{7} & -\frac{6}{7} & -\frac{3}{7} \\ \frac{5}{7} & \frac{2}{7} & -\frac{9}{7} & \frac{6}{7} \end{bmatrix}$.

C. $X = \begin{bmatrix} \frac{22}{7} & \frac{5}{7} \\ \frac{13}{7} & \frac{2}{7} \\ -\frac{6}{7} & -\frac{9}{7} \\ -\frac{3}{7} & \frac{6}{7} \end{bmatrix}$. **D.** X does not exist. **E.** None of the others.

(**Question 4 to question 5**)

The population of a species is divided into 3 age classes: Class I: from 0 to 3 months old, Class II: from 3 to 6 months old and Class III: from 6 to 9 months old. Suppose that after each 3 months, each individual in Class I, Class II and Class III produces 1, 6 and 4 offsprings, respectively. The survival rates after each

3 months of Class I and Class II are 70% and 90%, respectively. Suppose that at the beginning there are 1000 individuals in Class I (there is no individual in Class II and Class III).

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Question 4 (L.O.1, L.O.2). The Leslie transition matrix is

A. None of the others.
B. $\begin{bmatrix} 0 & 6 & 4 \\ 0.7 & 0 & 0 \\ 0 & 0.9 & 0.7 \end{bmatrix}$.
C. $\begin{bmatrix} 6 & 10 & 6 \\ 0.8 & 0 & 0.7 \\ 0.9 & 0 & 0.1 \end{bmatrix}$.
D. $\begin{bmatrix} 1 & 6 & 4 \\ 0.7 & 0 & 0 \\ 0 & 0.9 & 0 \end{bmatrix}$.
E. $\begin{bmatrix} 1 & 7 & 5 \\ 0.8 & 0 & 0 \\ 0 & 0.9 & 0 \end{bmatrix}$.

Question 5 (L.O.1, L.O.2). After 1 year, how many individuals are there in Class I? (Round the answer to the nearest integer).

A. None of the others.
B. 47900.
C. 8344.
D. 36280.
E. 3276.

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(Question 6 to question 9)

The population of a city is divided into two subgroups: suburb (I) and center (II). According to a survey, after each year, the probability that a person moves from the suburb to the center is 0.14 and from the center to the suburb is 0.06. At the initial time, there are 2 millions of people living in the suburb and 6 millions of people living in the center. Suppose that the number of people that are born, die and move between cities is negligible.

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Question 6 (L.O.1, L.O.2). Find the Markov transition matrix.

A. $A = [[0.860.14][0.060.94]]$.
B. $A = [[0.940.14][0.060.86]]$.
C. $A = [[0.940.06][0.140.86]]$.
D. $A = [[0.860.06][0.140.94]]$.
E. None of the others.

Question 7 (L.O.1, L.O.2). After 2 years, how many people live in the center?

A. None of the others.
B. 5.86.
C. 2.08.
D. 5.57.
E. 2.14.

Question 8 (L.O.1, L.O.2). Suppose that the population distribution reaches equilibrium, find the percentage of citizens who live in the center (compared with the total population).

A. 40.00%.
B. None of the othersc.
C. 30.00%.
D. 50.00%.
E. 70.00%.

Question 9 (L.O.1, L.O.2). In the vector space $M_2(\mathbb{R})$ of all 2×2 real matrices, let $F = \{X \in M_2(\mathbb{R}) | AX = X\}$ be a subspace. One basis of F is

A. $\left\{\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\right\}$.
B. $\left\{\begin{bmatrix} 0.06 & 0 \\ 0.14 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0.06 \\ 0 & 0.14 \end{bmatrix}\right\}$.
C. $\{(6, 14)\}$.
D. $\left\{\begin{bmatrix} -14.0 & 6.0 \\ 14.0 & -6.0 \end{bmatrix}\right\}$.
E. None of the others.

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(Question 10 to question 11)

Let m be a real number; $x = (2, 1, 1)$, $u_1 = (1, 2, 3)$, $u_2 = (3, 1, 3)$ and $u_3 = (0, 1, m)\}$ be vectors in the vector space $\mathbb{R}_3$.

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Question 10 (L.O.1, L.O.2). Find all values of m such that $\{u_1, u_2, u_3\}$ is a basis of $\mathbb{R}_3$.

A. $m \neq 16/5$.
B. None of the others.
C. $m \neq 6/5$.
D. $m \neq 11/5$.
E. $m \neq 1/5$.

Question 11 (L.O.1, L.O.2). Find all values of m such that the vector x is a linear combination of $\{u_1, u_3\}$.

A. $m = \frac{5}{3}$.
B. $m = \frac{2}{3}$.
C. $m = \frac{8}{3}$.

D. None of the others. **E.** m does not exist.

Question 12 (L.O.1, L.O.2). In the vector space $\mathbb{R}_4$, let $F = span\{(1,1,-2,2),(2,1,-1,2),(-4,-1,-1,-2)\}$ be a subspace. Find one basis of F.

A. F has no basis. **B.** $\{(1,1,-2,2),(2,1,-1,2)\}$. **C.** None of the others.
D. $\{(1,1,-2,2),(0,-1,3,-2),(0,0,0,0)\}$. **E.** $\{(1,1,-2,2),(2,1,-1,2),(-4,-1,-1,-2)\}$.

(**Question 13 to question 14**)

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In $\mathbb{R}_3$, let $E = \{(1,1,2),(3,2,1),(2,1,-2)\}$ and $F = \{(2,1,1),(1,1,1),(-2,1,2)\}$ be two bases.

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Question 13 (L.O.1, L.O.2). Find the transition matrix from E to F (which is denoted as $P_{F\leftarrow E}$).

A. $P_{F\leftarrow E} = \begin{bmatrix} -2 & -1 & 1 \\ 12 & 7 & 0 \\ -7 & -4 & 0 \end{bmatrix}$. **B.** $P_{F\leftarrow E} = \begin{bmatrix} -3 & 2 & 16 \\ 3 & -1 & -12 \\ -2 & 1 & 9 \end{bmatrix}$. **C.** None of the others.

D. $P_{F\leftarrow E} = \begin{bmatrix} 3 & -2 & -8 \\ -3 & 5 & 12 \\ 1 & -1 & -3 \end{bmatrix}$. **E.** $P_{F\leftarrow E} = \begin{bmatrix} 0 & -4 & -7 \\ 0 & 7 & 12 \\ 1 & -1 & -2 \end{bmatrix}$.

Question 14 (L.O.1, L.O.2). Let $u \in \mathbb{R}_3$ be a vector whose coordinate vector with respect to the basis E is $[u]_E = \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix}$. Find $[u]_F$.

A. $[u]_F = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}$. **B.** $[u]_F = \begin{bmatrix} -40 \\ 69 \\ -13 \end{bmatrix}$. **C.** $[u]_F = \begin{bmatrix} 76 \\ -57 \\ 43 \end{bmatrix}$.

D. None of the others. **E.** $[u]_F = \begin{bmatrix} -44 \\ 69 \\ -17 \end{bmatrix}$.

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(**Question 15 to question 20**)

Let $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 1 & 3 & -1 \\ 0 & 1 & 3 & 2 \\ 5 & m & 4 & -3 \end{bmatrix}$, $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in M_{4\times 1}(\mathbb{R})$ and $b = \begin{bmatrix} 5 \\ 6 \\ 7 \\ 10 \end{bmatrix}$.

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Question 15 (L.O.1, L.O.2). Find all real values of m such that A is invertible.

A. $m \neq 1$. **B.** $m \neq 5$. **C.** $m \neq 4$.
D. $m \neq 3$. **E.** None of the others.

Question 16 (L.O.1, L.O.2). Find the trace of $A \cdot A^T$

A. None of the others.
B. $trace(A \cdot A^T) = m^2 + 83$.
C. $trace(A \cdot A^T) = 3m^2 + 241$.
D. $trace(A \cdot A^T) = m^2 + 86$.
E. $trace(A \cdot A^T) = 2m^2 + 161$.

Question 17 (L.O.1, L.O.2). Let $m = 1$ and $B \in M_4(\mathbb{R})$ be a matrix, where $\det(B) = -2$. Find $\det(2A^T \cdot (B^3)^{-1})$.

A. None of the others. **B.** $\det(2A^T \cdot (B^3)^{-1}) = 22$. **C.** $\det(2A^T \cdot (B^3)^{-1}) = -11$.
D. $\det(2A^T \cdot (B^3)^{-1}) = -\frac{11}{4}$. **E.** $\det(2A^T \cdot (B^3)^{-1}) = \frac{11}{2}$.

Question 18 (L.O.1, L.O.2). Let $m = 2$. Which of the following statements about the linear system $AX = b$ is CORRECT?

A. The system has 4 solutions.
B. The system has a unique solution.

C. The system has infinitely many solutions.
D. The system has no solution.
E. None of the others.

Question 19 (L.O.1, L.O.2). With $m = 2$, let $X = \{x \in \mathbb{R}^4 | Ax = 0\}$ be a subspace of $\mathbb{R}^4$. Find one basis of X.

A. $\{(1, 1, -3, 1), (2, 1, 0, 1))\}$. **B**. $\{(1, 1, -3, 2), (\frac{3}{2}, -\frac{11}{4}, \frac{1}{4}, 1)\}$. **C**. X has no basis.
D. $\{(\frac{3}{2}, -\frac{11}{4}, \frac{1}{4}, 1)\}$. **E**. None of the others.

Question 20 (L.O.1, L.O.2). With $m = 2$, consider the linear system $AX = b$ as in the question 18. Find all solutions of the above system.

A. None of the others.
B. $(\frac{3t}{2} + 2, 2 - \frac{15t}{4}, t + 3, 2), t \in \mathbb{R}$.
C. $(\frac{3t}{2} + 1, -\frac{11t}{4}, \frac{9t}{4} + 2, t), t \in \mathbb{R}$.
D. $(\frac{3t}{2} + 1, 2 - \frac{11t}{4}, \frac{t}{4} + 1, t + 1), t \in \mathbb{R}$.
E. $(\frac{3t}{2} + 2, 2 - \frac{15t}{4}, \frac{5t}{4} + 2, 2t + 1), t \in \mathbb{R}$.

==================== The end ====================

RĐ:Đậu Thế Phiệt	Ngày:	PD:Nguyễn Tiến Dũng	Ngày ....
Ký tên		Ký tên	

Đại học Bách khoa-ĐHQG TPHCM Khoa Khoa học Ứng dụng	**THI GIỮA KỲ**	**Kỳ/năm học**	**II** | **2023-2024**
		Ngày thi	**04/08/2024**
	Môn học	**Môn ĐẠI SỐ TUYẾN TINH**	
	Mã môn học	**MT1007**	
	Thời gian	50 phút	Mã đề | 1102

Notes: - Đề thi trắc nghiệm gồm có 40 câu/4 trang.
- *Sinh viên không được dùng tài liệu. Nộp lại đề thi và giấy nháp cho giám thị.*
-*Mỗi câu trắc nghiệm sai: -1/5 số điểm của câu đó. Nếu không khoanh thì không trừ điểm.*

EXAM

(**Question 1 to question 3**)

Let $A = \begin{bmatrix} -3 & 1 & -2 & 1 \\ -2 & 2 & 1 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & m \\ -3 & -2 \end{bmatrix}$, $m \in \mathbb{R}$, be two matrices.

Question 1 (L.O.1, L.O.2). Which of the following statements is CORRECT?

A. BA does not exist.

B. $AB = \begin{bmatrix} 3 & 4-3m \\ -5 & m-4 \\ -5 & -2m-2 \\ 7 & m+4 \end{bmatrix}$.

C. $A + B = \begin{bmatrix} -2 & m+1 \\ -5 & 0 \end{bmatrix}$.

D. $BA = \begin{bmatrix} -2m-3 & 2m+1 & m-2 & 1-2m \\ 13 & -7 & 4 & 1 \end{bmatrix}$

E. None of the others.

Question 2 (L.O.1, L.O.2). Let $f(x) = 3x^2 - 2x - 4$ be a polynomial. Find $f(B)$.

A. $f(B) = \begin{bmatrix} -9m-3 & -5m \\ 15 & 12-9m \end{bmatrix}$.

B. $f(B) = \begin{bmatrix} -6m-5 & -5m-4 \\ 11 & 10-6m \end{bmatrix}$.

C. None of the others.

D. $f(B) = \begin{bmatrix} -6m-5 & 9-2m \\ 6-3m & 10-6m \end{bmatrix}$.

E. $f(B) = \begin{bmatrix} -6m-5 & 5-2m \\ 2-3m & 10-6m \end{bmatrix}$.

Question 3 (L.O.1, L.O.2). Given that $m = 2$. Find a matrix X such that: $2BX = 3A + X$.

A. None of the others.

B. $X = \begin{bmatrix} -\frac{34}{19} & \frac{6}{19} & -\frac{32}{19} & \frac{22}{19} \\ -\frac{58}{19} & \frac{46}{19} & \frac{8}{19} & -\frac{34}{19} \end{bmatrix}$.

C. $X = \begin{bmatrix} \frac{9}{19} & \frac{30}{19} \\ \frac{21}{19} & -\frac{6}{19} \\ \frac{48}{19} & \frac{27}{19} \\ -\frac{51}{19} & -\frac{18}{19} \end{bmatrix}$.

D. X does not exist.

E. $X = \begin{bmatrix} \frac{69}{19} & -\frac{39}{19} & \frac{18}{19} & \frac{9}{19} \\ -\frac{60}{19} & \frac{24}{19} & -\frac{33}{19} & \frac{12}{19} \end{bmatrix}$.

ĐÁP ÁN đề 1101 - ngày thi 04/08/2024

1 A	**3** B	**5** D	**7** B	**9** B	**11** A	**13** D	**15** E	**17** B	**19** D
2 E	**4** D	**6** D	**8** E	**10** C	**12** B	**14** E	**16** B	**18** C	**20** D

ĐÁP ÁN đề 1102 - ngày thi 04/08/2024

1 D	**3** E	**5** D	**7** D	**9** C	**11** A	**13** B	**15** C	**17** A	**19** A
2 A	**4** B	**6** A	**8** C	**10** D	**12** B	**14** B	**16** B	**18** A	**20** A

ĐÁP ÁN đề 1103 - ngày thi 04/08/2024

1 B	**3** B	**5** E	**7** C	**9** B	**11** A	**13** C	**15** A	**17** C	**19** E
2 A	**4** A	**6** A	**8** C	**10** B	**12** D	**14** B	**16** E	**18** E	**20** B

ĐÁP ÁN đề 1104 - ngày thi 04/08/2024

1 E	**3** C	**5** C	**7** E	**9** D	**11** C	**13** A	**15** C	**17** B	**19** E
2 D	**4** B	**6** D	**8** E	**10** C	**12** C	**14** E	**16** A	**18** D	**20** A

ĐÁP ÁN đề 2201 - ngày thi 04/08/2024

1 A	**3** E	**5** E	**7** A	**9** E	**11** A	**13** C	**15** C	**17** B	**19** B
2 A	**4** D	**6** D	**8** D	**10** C	**12** E	**14** A	**16** E	**18** B	**20** A

ĐÁP ÁN đề 2202 - ngày thi 04/08/2024

1 D	**3** A	**5** D	**7** B	**9** C	**11** A	**13** E	**15** B	**17** E	**19** A
2 B	**4** E	**6** E	**8** D	**10** C	**12** A	**14** A	**16** A	**18** A	**20** C

ĐÁP ÁN đề 2203 - ngày thi 04/08/2024

1 D	**3** A	**5** D	**7** C	**9** E	**11** E	**13** A	**15** C	**17** D	**19** C
2 D	**4** E	**6** D	**8** A	**10** C	**12** C	**14** B	**16** C	**18** C	**20** B

ĐÁP ÁN đề 2204 - ngày thi 04/08/2024

1 E	**3** D	**5** C	**7** C	**9** C	**11** D	**13** B	**15** E	**17** D	**19** B
2 B	**4** A	**6** B	**8** E	**10** C	**12** A	**14** A	**16** E	**18** A	**20** A